FOREWORD

The first man-made object was sent into a low orbit around the earth in 1957, just three years after I was born. I was a schoolboy when the first humans walked on the surface of the moon. Since then the exploration of space has continued steadily and instruments of increasing complexity have been launched into space. One of these, Voyager 1, has even left our solar system. The information gathered has revolutionized our understanding of space, the birth and death of stars, and has widened our horizons to the point where we have proved the existence of many planets that might be similar to earth, orbiting stars in far-flung galaxies. It is hard not be impressed at what we have achieved in such a short time. For a little over 400 years since Galileo first saw the moons of Jupiter, we have been looking outward—but in my lifetime we have been able to look back at ourselves. For me, one single image from all this monumental effort stands out and it is not one of a Martian landscape or the icy surface of a comet, but the now famous “blue marble” photograph of the earth, taken on December 7th, 1972, by the crew of the Apollo 17 spacecraft, at a distance of about 45,000 kilometers. It shows us how small our home really is and, more importantly, that it is finite.

We know a lot about the physical composition of the earth and through studying rocks and the fossil record we have been able to piece together the history of life. We know that evolution has produced, as Darwin eloquently put it, “endless forms most beautiful and most wonderful,” but we are still a very long way off putting a number on exactly how many species we share the earth with. What we do know for sure is that relatively few of them, just shy of 3 percent, have a backbone. Most species and the majority of animals are invertebrates with 3 or more pairs of legs—insects, spiders, and crustaceans. Some estimates of global species numbers have been set as high as 30 million or more, but a general consensus view is that it may be somewhere between 8 and 12 million species. Despite our increasing mastery of biology, from discovering and deciphering the code of life to the making of synthetic organisms, it is very unlikely that we will ever name and catalog the full extent of life on our planet. We must, therefore, get used to the idea that the majority of species will come and go without us ever knowing they existed at all.